

Multicomponent Reactions

Iron-Nitrate-Catalyzed Oxidative Esterification of Aldehydes and Alcohols with *N*-Hydroxyphthalimide: Efficient Synthesis of *N*-Hydroxyimide Esters

Xiaohe Xu,^[a] Jian Sun,^[a] Yuyan Lin,^[a] Jingya Cheng,^[a] Pingping Li,^[a] Xiaoying Jiang,^[a] Renren Bai,^[a] and Yuanyuan Xie*^[a,b]

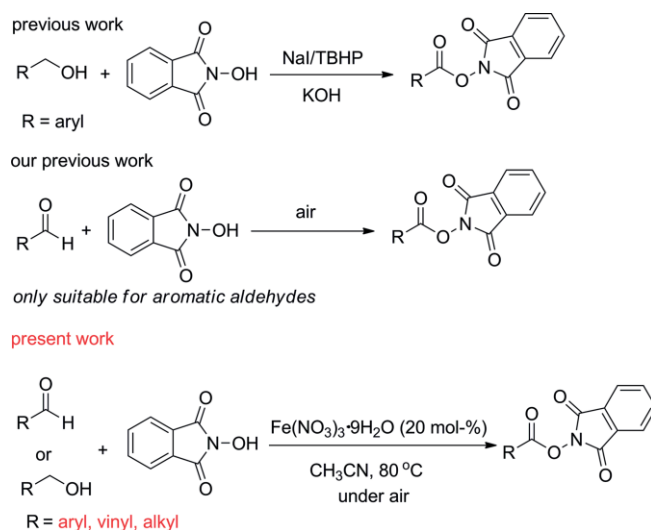
Abstract: An $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ -catalyzed cross-dehydrogenative coupling reaction between *N*-hydroxyphthalimide (NHPI) or *N*-hydroxysuccinimide (NHSI) and aldehydes or alcohols in air is

described. This transformation represents an efficient approach to the preparation of *N*-hydroxyimide ester derivatives in moderate to excellent yields, and has a wide substrate scope.

Introduction

Ester groups are important functional groups in organic chemistry. They can be found in natural products, medicinally relevant molecules, synthetic materials, and polymers.^[1] Therefore, many methods have been developed for their efficient construction.^[2] *N*-Hydroxyimide esters, an important class of esters, are reactive intermediates that are used in the preparation of amides, esters, and *O*-substituted hydroxylamines, and in the formation of $\text{C}(\text{sp}^2)\text{--C}(\text{sp}^2)$ bonds by cross-coupling.^[3] Although *N*-hydroxyimide esters are frequently used intermediates, few approaches have been developed for their preparation. Traditionally, *N*-hydroxyamide esters were synthesized by the direct coupling of carboxylic acids with *N*-hydroxyimides in the presence of *N,N'*-dicyclohexylcarbodiimide.^[4] This approach suffers from several drawbacks, such as poor atom efficiency and difficult purification. The cross-dehydrogenative coupling (CDC) approach is attractive, because it is atom-economic but it does not require prefunctionalization of the substrates.^[5] Recently, several groups have reported the formation of *N*-hydroxyamide esters from aldehydes via oxidative cross-dehydrogenative coupling, either in the presence of a metal catalyst or under metal-free conditions.^[6] These reported cross-dehydrogenative coupling reactions between *N*-hydroxyphthalimide (NHPI) and aldehydes were normally applicable to aromatic aldehydes. The CDC reaction between NHPI and alkyl aldehydes, α,β -unsaturated aldehydes, or alcohols still remains a major challenge. In 2013, Yu and coworkers developed a sodium-iodide-catalyzed cross-coupling reaction of alcohols with NHPI in the presence

of TBHP (*tert*-butylhydroperoxide) and KOH (Scheme 1).^[7] Although the above-mentioned elegant methods appear to be general and efficient, they still suffer from some drawbacks, such as the use of stoichiometric oxidants and heavy-metal catalysts. Hence, milder, more environmentally benign synthetic methods with a wider substrate scope are still required for the synthesis of *N*-hydroxyamide esters.



Scheme 1. Various approaches to the synthesis of *N*-hydroxyimide esters.

Iron is the second-most abundant metal in the earth's crust.^[8] Compared with other metals, iron is relatively nontoxic, inexpensive, and environmentally friendly. Thus a variety of organic transformations have been carried out using iron-based catalyst systems.^[9] Recently, iron(III) nitrate nonahydrate [$\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$] has been used as a nitration reagent; it has also been widely used as a catalyst for the oxidation of benzyl alcohols, and for the oxidative coupling of NHPI with olefins.^[10] It is well known that the thermal decomposition of $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ generates nitrogen dioxide (NO_2),^[11] a radical species. In the presence of NO_2 , NHPI can be oxidized to PINO (phthal-

[a] Collaborative Innovation Center of Yangtze River Delta Region Green Pharmaceuticals, Zhejiang University of Technology, Hangzhou, P. R. China

[b] College of Pharmaceutical Sciences, Zhejiang University of Technology, Hangzhou, P. R. China
E-mail: xyycz@zjut.edu.cn
<https://renrenbai.wixsite.com/researchgroup>

Supporting information for this article is available on the WWW under <https://doi.org/10.1002/ejoc.201701411>.

imide *N*-oxyl), which may initiate radical oxidative reactions.^[12] In this paper, we report an example of an $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ -catalyzed intermolecular C–O cross-coupling reaction of aldehydes or alcohols with NHPI.

Results and Discussion

For our initial study, propionaldehyde (**1a**) and NHPI (**2a**) were chosen as model substrates. The cross-dehydrogenative coupling reaction was run in CH_3CN at room temperature in the presence of $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ under air for 1 h, and the desired *N*-hydroxyamide ester **3aa** was obtained in 13 % yield (Table 1, entry 1). When the reaction was run at elevated temperature, the desired product was obtained in a much better yield (Table 1, entry 2). When the temperature was increased to 80 °C, the starting material NHPI was completely converted into the desired product (Table 1, entry 3). Other catalysts, including $\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$, $\text{Mg}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$, and AgNO_3 , were also tested, but they were less active, and gave **3aa** in 72, 59, and 50 % yields, respectively (Table 1, entries 4–6). The reaction did not give the coupling product when other iron salts such as $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$ and $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$ were used (Table 1, entries 7 and 8); this result indicates the importance of NO_3^- in the reaction. However, the reaction did not take place in the presence of NaNO_2 (Table 1, entry 9). It has been reported that NO_2 can be generated from NaNO_2 under acidic conditions.^[13] With this in mind, CH_3COOH (3 equiv.) was added to the reaction mixture, and **3aa** was obtained in 55 % yield (Table 1, entry 9). The effect of different solvents, including EtOAc, acetone, toluene, and

DMF, was also investigated, and CH_3CN was found to be the optimal solvent (Table 1, entries 10–13). Decreasing the loading of $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ lowered the yield. When the reaction time was increased to 4 h, the yield of **3aa** increased to 76 %. However, when the reaction time was prolonged further, the yield of **3aa** was unchanged (Table 1, entry 14). The reaction did not take place in the absence of $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (Table 1, entry 15), and when the reaction was carried out under a nitrogen atmosphere, the yield of **3aa** was only 23 % (entry 16). These results show that both a catalyst and an oxidant (air) are crucial for this transformation. Hence, the optimal conditions were determined to be: **1a** (1.0 mmol) with **2a** (0.5 mmol) in the presence of $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (0.1 mmol) in CH_3CN at 80 °C under air. This gave the desired product in 82 % yield (Table 1, entry 3).

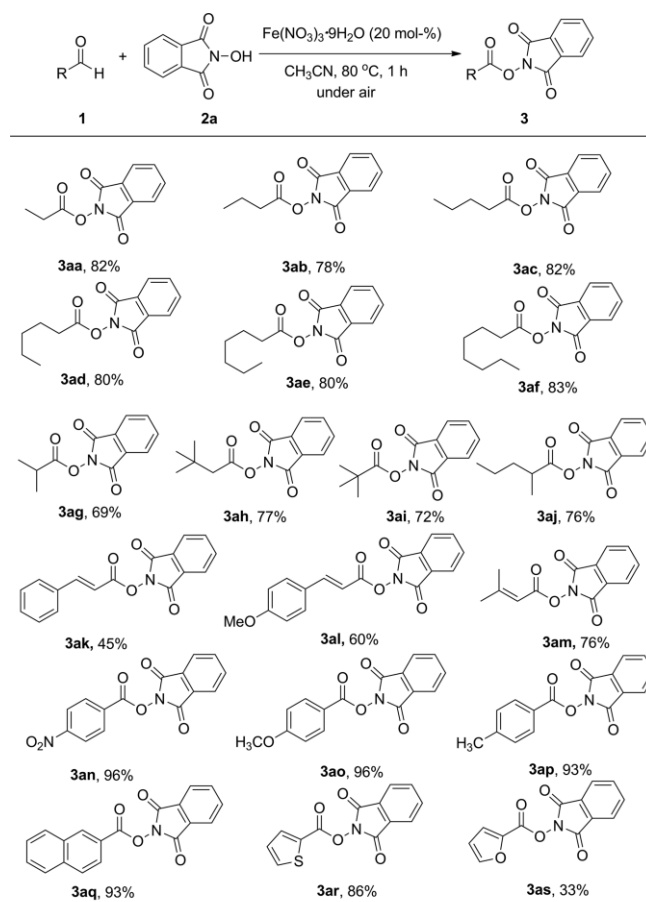
Having established these optimized reaction conditions, we then investigated the scope of the aldehyde substrates. As shown in Table 2, a series of aldehydes were found to react smoothly with NHPI, and the corresponding products were obtained in good to excellent yields. The aliphatic aldehydes propionaldehyde, butyraldehyde, and pentanaldehyde delivered the corresponding products **3aa**, **3ab**, and **3ac** in 82, 78, and 82 % yields, respectively. Increasing the chain length had little effect on the efficiency of the reaction, and products **3ad–3af**

Table 1. Optimization of the reaction conditions.^[a]

Entry	Catalyst	<i>T</i> [°C]	Solvent	Yield [%] ^[b]
1	$\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$	room temp.	CH_3CN	13
2	$\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$	60	CH_3CN	68
3	$\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$	80	CH_3CN	82
4	$\text{Cu}(\text{NO}_3)_2 \cdot 3\text{H}_2\text{O}$	80	CH_3CN	72
5	$\text{Mg}(\text{NO}_3)_2 \cdot 6\text{H}_2\text{O}$	80	CH_3CN	59
6	AgNO_3	80	CH_3CN	50
7	$\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$	80	CH_3CN	n.d. ^[c]
8	$\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$	80	CH_3CN	n.d. ^[c]
9	NaNO_2	80	CH_3CN	trace (55) ^[d]
10	$\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$	77	EtOAc	46
11	$\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$	56	acetone	35
12	$\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$	80	toluene	13
13	$\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$	80	DMF	30
14 ^[e]	$\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$	80	CH_3CN	65 (76) ^[f]
15 ^[g]	–	80	CH_3CN	n.d. ^[c]
16 ^[h]	$\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$	80	CH_3CN	23

[a] Unless otherwise stated, reactions were carried out with **1a** (1.0 mmol), **2a** (0.5 mmol), catalyst (20 mol-%), solvent (6.0 mL) under air for 1 h. [b] Isolated yields based on **2a**. [c] n.d. = not detected. [d] CH_3COOH (3.0 equiv.) was added. [e] 10 mol-% $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ was used. [f] The reaction time was 4 h. [g] $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ was not added. [h] The reaction was carried out under a nitrogen atmosphere.

Table 2. Scope of the reaction of aldehydes with NHPI.^[a,b]

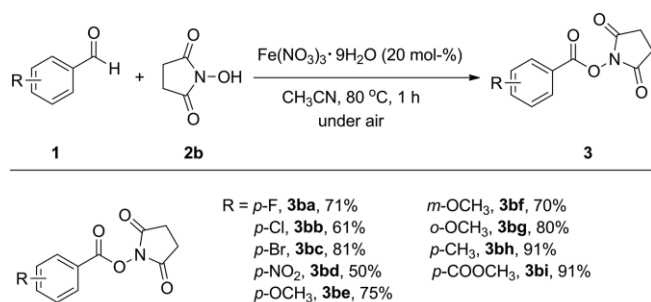


[a] All reactions were carried out with **1** (1 mmol), **2a** (0.5 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (0.1 mmol), CH_3CN (6.0 mL) at 80 °C under air for 1 h. [b] Isolated yields based on **2a**.

were obtained in 80–83 % yield. Due to steric effects, the branched aliphatic aldehydes reacted less efficiently. For example, when isobutyraldehyde was used as a substrate, the cross-dehydrogenative coupling product **3ag** was obtained in only 69 % yield. Other branched aliphatic aldehydes such as 3,3-dimethylbutanal, pivalaldehyde, and 2-methylpentanal gave lower yields, and products **3ah**, **3ai**, and **3aj** were obtained in 77, 72, and 76 %, respectively. α,β -Unsaturated aldehydes were found to be suitable substrates, and esters **3ak–3am** were obtained in 45–76 % yield. This work provides the first example of a direct transformation of alkyl α,β -unsaturated aldehydes and NHPI to give the corresponding *N*-hydroxyimide esters. The optimized conditions were also compatible with benzaldehyde derivatives and 2-naphthaldehyde, which gave the desired products **3an–3aq** in 93–96 % yield. Heterocyclic aldehydes, including thiophene-2-carbaldehyde and furfural, were also found to be compatible with the reaction conditions. For instance, 1,3-dioxoisindolin-2-yl thiophene-2-carboxylate (**3ar**) was produced in 86 % yield, and 1,3-dioxoisindolin-2-yl furan-2-carboxylate (**3as**) was formed in 33 % yield.

We then investigated the reaction of aldehydes with *N*-hydroxysuccinimide (**2b**; NHSI) under the optimized reaction conditions. The results are shown in Table 3. Aldehydes with electron-withdrawing groups performed well, and the desired products **3ba–3bd** were formed in 50–81 % yield. Other substituents on the benzene ring of the aldehyde, including methoxy, methyl, and ester groups, were all well tolerated and the corresponding products **3be–3bi** were obtained in good to excellent isolated yields. However, aliphatic aldehydes turned out to be poor substrates for this transformation.

Table 3. Scope of the reaction of aldehydes with NHSI.^[a,b]

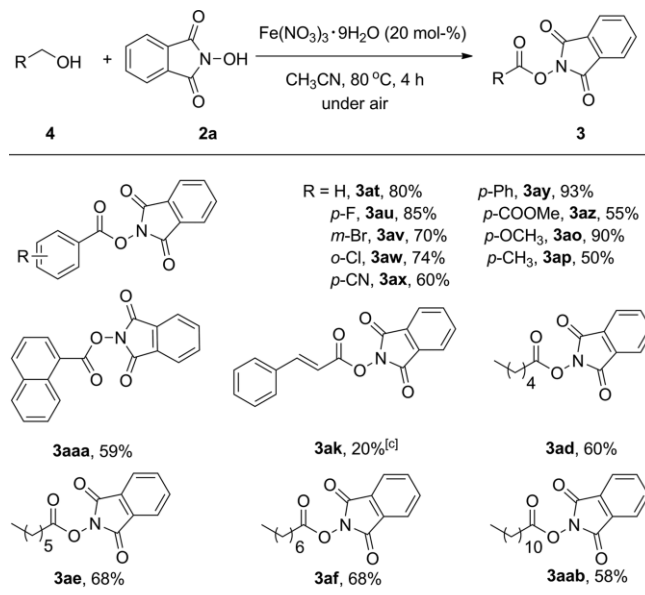


[a] All reactions were carried out with **1** (1 mmol), **2b** (0.5 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (0.1 mmol), CH_3CN (6.0 mL) at 80 °C under air for 1 h. [b] Isolated yields based on **2b**.

Encouraged by the results described above, we further investigated the possibility of using alcohols as coupling partners with NHPI, since alcohols can be readily oxidized to aldehydes by NHPI and $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$. As shown in Table 4, benzylic alcohols with various substituents, including electron-withdrawing groups such as F, Cl, Br, CN, Ph, and COOMe, or electron-donating groups such as OCH_3 and CH_3 , were converted into the desired products in moderate to excellent yields. Naphthalen-1-ylmethanol and 3-phenylprop-2-en-1-ol also gave the desired products **3aaa** and **3ak** in 59 and 20 % yields, respectively. Ali-

phatic alcohols **4d–4f** and **4b** were investigated under the optimized conditions, and the corresponding products **3ad–3af** and **3aab** were obtained, albeit in moderate yields (58–68 %).

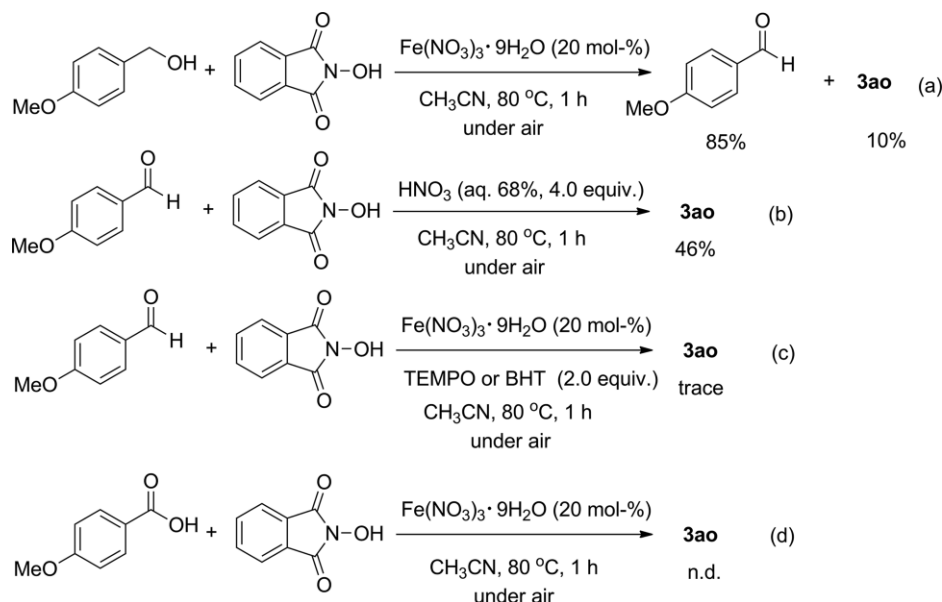
Table 4. Scope of the reaction of alcohols with NHPI.^[a,b]



[a] All reactions were carried out with **1** (1 mmol), **2b** (0.5 mmol), $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (0.1 mmol), CH_3CN (6.0 mL) at 80 °C under air for 4 h unless otherwise stated. [b] Isolated yields based on **2a**. [c] The reaction was carried out for 24 h.

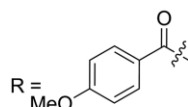
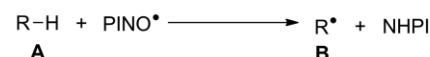
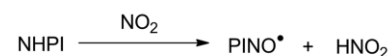
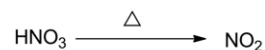
Having established these methods, several control experiments were carried out to probe the reaction mechanism (Scheme 2). Under the standard conditions, *p*-methoxybenzyl alcohol was oxidized to *p*-methoxybenzaldehyde in 85 % yield in 1 h (Scheme 2, a). When the reaction was carried out using HNO_3 , the desired product **3ao** was obtained in 46 % yield (Scheme 2, b). The esterification of 4-methoxybenzaldehyde (**1o**) with NHPI (**2a**) was carried out in the presence of TEMPO [(2,2,6,6-tetramethylpiperidin-1-yl)oxyl] under the standard reaction conditions, and only traces of the product **3ao** were obtained (Scheme 2, c). The model reaction was also completely inhibited by another radical scavenger, 2,6-di-*tert*-butyl-4-methylphenol (BHT) (Scheme 2, c). To check whether *p*-methoxybenzoic acid was formed in situ as an active intermediate, *p*-methoxybenzoic acid was subjected to the cross-coupling reaction with NHPI (Scheme 2, d), and *N*-hydroxyimide ester **3ao** was not formed under the standard conditions.

Based on the results of the above experiments taken together with previous reports,^[6,10–12] a plausible reaction pathway is shown in Scheme 3. Initially, ferric nitrate is hydrolyzed to give HNO_3 , and then thermal decomposition of HNO_3 generates NO_2 . In the gas phase of the reaction mixture, the presence of NO_2 was confirmed by its characteristic reddish-brown color. In the presence of NO_2 , NHPI forms the PINO radical, which induces homolysis of the C–H bond of the aldehyde to give acyl radical **B**.^[14] Finally, recombination of acyl radical **B** and



Scheme 2. Control experiments.

the PINO radical forms *N*-hydroxyimide ester **3ao**. The HNO_2 decomposes to give HNO_3 and NO . In the presence of O_2 , NO is oxidized to NO_2 , and the NO_2 is recycled.



Scheme 3. Plausible mechanism.

Conclusions

We have described an efficient method for the synthesis of *N*-hydroxyimide esters. The notable features of this reaction are a broad substrate scope, and environmentally benign conditions. The use of an inexpensive, nontoxic catalyst and a green oxidant (air) makes this protocol attractive for synthetic chemistry. This is also the first report that describes the use of both alcohols and aldehydes (including alkyl aldehydes and α,β -unsaturated aldehydes) as the coupling partners for NHPI to give the corresponding esters. Further studies on the coupling of other compounds with *N*-hydroxyimides are underway.

Experimental Section

General Procedure for the Synthesis of **3aa:** Propionaldehyde (**1a**; 1.0 mmol), NHPI (0.5 mmol, 1.0 equiv.), and $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (0.1 mmol) were dissolved in CH_3CN (6.0 mL), and the mixture was stirred for 1 h at 80 °C under air. After this time, the NHPI was completely consumed. Then, the solvent was removed under reduced pressure. The residue was washed with saturated NaHCO_3 , and then extracted with EtOAc ($2 \times 20 \text{ mL}$). The combined organic layers were dried with anhydrous Na_2SO_4 , and concentrated. The residue was purified by column chromatography on silica gel (petroleum ether/ EtOAc , 10:1) to give the desired product **3aa** (90 mg, 82 %) as a white solid. M.p. 88–90 °C. ^{63}C ^1H NMR (600 MHz, CDCl_3): δ = 7.90–7.88 (m, 2 H), 7.81–7.79 (m, 2 H), 2.46 (q, J = 7.8 Hz, 2 H), 1.32 (t, J = 7.2 Hz, 3 H) ppm. ^{13}C NMR (125 MHz, CDCl_3): δ = 170.4, 162.0, 134.8, 128.9, 124.0, 24.5, 8.7 ppm. IR (KBr): $\tilde{\nu}$ = 3433, 3097, 2981, 2936, 1755, 1736, 1610, 1185, 1137, 1104, 1059, 1000, 876, 699, 520 cm^{-1} . HRMS (ESI): calcd. for $[\text{M} + \text{Na}]^+$ $\text{C}_{11}\text{H}_9\text{NNaO}_4$ 242.0424; found 242.0423.

1,3-Dioxoisindolin-2-yl Butyrate (3ab**):** Oil. ^{63}C ^1H NMR (600 MHz, CDCl_3): δ = 7.90–7.87 (m, 2 H), 7.81–7.78 (m, 2 H), 2.66 (t, J = 7.2 Hz, 2 H), 1.86–1.80 (m, 2 H), 1.08 (t, J = 7.2 Hz, 3 H) ppm. ^{13}C NMR (125 MHz, CDCl_3): δ = 169.5, 162.0, 134.7, 128.9, 123.9, 32.8, 18.3, 13.4 ppm. IR (KBr): $\tilde{\nu}$ = 3521, 3068, 2969, 2937, 1788, 1743, 1609, 1467, 1367, 1186, 1127, 1062, 969, 877, 787, 696, 518 cm^{-1} . HRMS (ESI): calcd. for $[\text{M} + \text{Na}]^+$ $\text{C}_{12}\text{H}_{11}\text{NNaO}_4$ 256.0580; found 256.0591.

1,3-Dioxoisindolin-2-yl Pentanoate (3ac**):** Oil. ^1H NMR (600 MHz, CDCl_3): δ = 7.89–7.87 (m, 2 H), 7.79–7.78 (m, 2 H), 2.67 (t, J = 7.8 Hz, 2 H), 1.80–1.75 (m, 2 H), 1.51–1.45 (m, 2 H), 0.97 (t, J = 7.2 Hz, 3 H) ppm. ^{13}C NMR (125 MHz, CDCl_3): δ = 169.6, 162.0, 134.7, 129.0, 123.9, 30.7, 26.7, 22.0, 13.6 ppm. IR (KBr): $\tilde{\nu}$ = 3521, 3045, 2960, 2932, 1788, 1743, 1670, 1467, 1364, 1186, 1128, 1064, 878, 696, 518 cm^{-1} . HRMS (ESI): calcd. for $[\text{M} + \text{Na}]^+$ $\text{C}_{13}\text{H}_{13}\text{NNaO}_4$ 270.0737; found 270.0734.

1,3-Dioxoisindolin-2-yl Hexanoate (3ad**):** Oil. ^1H NMR (600 MHz, CDCl_3): δ = 7.91–7.90 (m, 2 H), 7.81–7.80 (m, 2 H), 2.68 (t, J = 7.2 Hz, 2 H), 1.84–1.79 (m, 2 H), 1.48–1.35 (m, 4 H), 0.95 (t, J = 7.2 Hz, 3 H)

ppm. ^{13}C NMR (125 MHz, CDCl_3): δ = 169.7, 162.0, 134.7, 129.0, 123.9, 31.0, 24.4, 22.2, 13.8 ppm. IR (KBr): $\tilde{\nu}$ = 3059, 3031, 2957, 2931, 1789, 1744, 1467, 1363, 1186, 1311, 1066, 966, 878, 786, 697, 518 cm^{-1} . HRMS (ESI): calcd. for $[\text{M} + \text{Na}]^+$ $\text{C}_{14}\text{H}_{15}\text{NNaO}_4$ 284.0893; found 284.0888.

1,3-Dioxoisindolin-2-yl Heptanoate (3ae): Oil. ^1H NMR (600 MHz, CDCl_3): δ = 7.89–7.87 (m, 2 H), 7.80–7.78 (m, 2 H), 2.67 (t, J = 7.2 Hz, 2 H), 1.81–1.76 (m, 2 H), 1.47–1.43 (m, 2 H), 1.35–1.33 (m, 4 H), 0.91 (t, J = 7.2 Hz, 3 H) ppm. ^{13}C NMR (125 MHz, CDCl_3): δ = 169.7, 162.0, 134.7, 128.9, 123.9, 31.3, 31.0, 28.5, 24.6, 22.4, 14.0 ppm. IR (KBr): $\tilde{\nu}$ = 3448, 3043, 3025, 2954, 2869, 1788, 1740, 1638, 1377, 1139, 1081, 877, 699, 521 cm^{-1} . HRMS (ESI): calcd. for $[\text{M} + \text{Na}]^+$ $\text{C}_{15}\text{H}_{17}\text{NNaO}_4$ 298.1050; found 298.1041.

1,3-Dioxoisindolin-2-yl Octanoate (3af): Oil. ^1H NMR (600 MHz, CDCl_3): δ = 7.90–7.86 (m, 2 H), 7.80–7.77 (m, 2 H), 2.67–2.65 (m, 2 H), 1.801–1.76 (m, 2 H), 1.43 (s, 2 H), 1.33–1.30 (m, 6 H), 0.90 (t, J = 3.6 Hz, 3 H) ppm. ^{13}C NMR (125 MHz, CDCl_3): δ = 169.7, 162.0, 134.7, 129.0, 123.9, 31.6, 31.0, 28.8, 24.7, 22.6, 14.1 ppm. IR (KBr): $\tilde{\nu}$ = 3065, 2958, 2928, 1789, 1745, 1467, 1364, 1185, 1072, 969, 878, 697, 518 cm^{-1} . HRMS (ESI): calcd. for $[\text{M} + \text{Na}]^+$ $\text{C}_{16}\text{H}_{19}\text{NNaO}_4$ 312.1206; found 312.1202.

1,3-Dioxoisindolin-2-yl Isobutyrate (3ag): White solid. M.p. 60–61 $^{\circ}\text{C}$. ^1H NMR (600 MHz, CDCl_3): δ = 7.89–7.88 (m, 2 H), 7.79–7.78 (m, 2 H), 2.99–2.94 (m, 1 H), 1.38 (d, J = 7.2 Hz, 6 H) ppm. ^{13}C NMR (125 MHz, CDCl_3): δ = 173.1, 162.0, 134.7, 129.0, 123.9, 31.8, 18.8 ppm. IR (KBr): $\tilde{\nu}$ = 3447, 3084, 2985, 2925, 1782, 1740, 1466, 1187, 1053, 967, 876, 694, 520 cm^{-1} . HRMS (ESI): calcd. for $[\text{M} + \text{Na}]^+$ $\text{C}_{12}\text{H}_{11}\text{NNaO}_4$ 256.0580; found 256.0575.

1,3-Dioxoisindolin-2-yl 3,3-Dimethylbutanoate (3ah): White solid. M.p. 92–93 $^{\circ}\text{C}$. ^1H NMR (600 MHz, CDCl_3): δ = 7.89–7.87 (m, 2 H), 7.79–7.78 (m, 2 H), 2.53 (s, 2 H), 1.17 (s, 9 H) ppm. ^{13}C NMR (125 MHz, CDCl_3): δ = 167.9, 162.1, 134.7, 129.0, 123.9, 44.6, 31.3, 29.5 ppm. IR (KBr): $\tilde{\nu}$ = 3428, 3072, 2963, 2926, 1782, 1736, 1468, 1370, 1286, 1182, 1090, 970, 878, 702, 519 cm^{-1} . HRMS (ESI): calcd. for $[\text{M} + \text{Na}]^+$ $\text{C}_{14}\text{H}_{15}\text{NNaO}_4$ 284.0893; found 284.0892.

1,3-Dioxoisindolin-2-yl Pivalate (3ai): Oil. ^1H NMR (600 MHz, CDCl_3): δ = 7.89–7.87 (m, 2 H), 7.79–7.77 (m, 2 H), 1.44 (s, 9 H) ppm. ^{13}C NMR (125 MHz, CDCl_3): δ = 174.4, 162.1, 134.7, 129.1, 123.9, 38.4, 27.0 ppm. IR (KBr): $\tilde{\nu}$ = 3521, 3066, 3031, 2977, 2936, 1758, 1744, 1481, 1370, 1186, 1062, 1023, 965, 878, 696, 518 cm^{-1} . HRMS (ESI): calcd. for $[\text{M} + \text{Na}]^+$ $\text{C}_{13}\text{H}_{13}\text{NNaO}_4$ 270.0737; found 270.0731.

1,3-Dioxoisindolin-2-yl 2-Methylpentanoate (3aj): Oil. ^1H NMR (600 MHz, CDCl_3): δ = 7.89–7.87 (m, 2 H), 7.79–7.78 (m, 2 H), 2.88–2.82 (m, 1 H), 1.86–1.80 (m, 1 H), 1.61–1.57 (m, 1 H), 1.53–1.48 (m, 2 H), 1.35 (d, J = 6.6 Hz, 3 H), 0.98 (t, J = 7.2 Hz, 3 H) ppm. ^{13}C NMR (125 MHz, CDCl_3): δ = 172.8, 162.1, 134.7, 129.0, 123.9, 36.9, 35.8, 20.1, 16.9, 13.9 ppm. IR (KBr): $\tilde{\nu}$ = 3431, 3097, 2958, 1768, 1743, 1635, 1377, 1141, 1102, 1045, 978, 877, 699, 518 cm^{-1} . HRMS (ESI): calcd. for $[\text{M} + \text{Na}]^+$ $\text{C}_{14}\text{H}_{15}\text{NNaO}_4$ 284.0893; found 284.0885.

1,3-Dioxoisindolin-2-yl Cinnamate (3ak): White solid. M.p. 143–145 $^{\circ}\text{C}$. ^1H NMR (600 MHz, CDCl_3): δ = 7.99 (d, J = 16.2 Hz, 1 H), 7.95–7.93 (m, 2 H), 7.84–7.82 (m, 2 H), 7.62 (d, J = 6.6 Hz, 2 H), 7.51–7.45 (m, 3 H), 6.69 (d, J = 16.2 Hz, 1 H) ppm. ^{13}C NMR (125 MHz, CDCl_3): δ = 163.0, 162.1, 150.0, 134.8, 133.6, 131.6, 129.1, 129.0, 128.7, 124.0, 111.7 ppm. IR (KBr): $\tilde{\nu}$ = 3438, 3036, 2926, 1744, 1630, 1384, 1139, 980, 927, 876, 620, 532 cm^{-1} . HRMS (ESI): calcd. for $[\text{M} + \text{Na}]^+$ $\text{C}_{17}\text{H}_{17}\text{NNaO}_4$ 316.0580; found 316.0582.

1,3-Dioxoisindolin-2-yl 3-(4-Methoxyphenyl)acrylate (3al): Brown solid. M.p. 135–137 $^{\circ}\text{C}$. ^1H NMR (600 MHz, CDCl_3): δ = 7.95–7.93 (m, 3 H), 7.83–7.81 (m, 2 H), 7.57 (d, J = 9.0 Hz, 2 H), 6.97 (d,

J = 9.0 Hz, 2 H), 6.54 (d, J = 16.2 Hz, 1 H), 3.89 (s, 3 H) ppm. ^{13}C NMR (125 MHz, CDCl_3): δ = 163.3, 162.5, 162.2, 149.7, 134.7, 130.6, 129.1, 126.4, 124.0, 114.6, 108.8, 55.5 ppm. IR (KBr): $\tilde{\nu}$ = 3448, 3046, 2929, 1778, 1744, 1635, 1363, 1145, 987, 947, 878, 626, 528 cm^{-1} . HRMS (ESI): calcd. for $[\text{M} + \text{Na}]^+$ $\text{C}_{18}\text{H}_{13}\text{NNaO}_5$ 346.0690; found 346.0686.

1,3-Dioxoisindolin-2-yl 3-Methylbut-2-enoate (3am): White solid, 118–119 $^{\circ}\text{C}$. ^1H NMR (600 MHz, CDCl_3): δ = 7.90–7.88 (m, 2 H), 7.79–7.78 (m, 2 H), 6.00 (t, J = 1.2 Hz, 1 H), 2.23 (d, J = 1.2 Hz, 3 H), 2.05 (d, J = 0.6 Hz, 3 H) ppm. ^{13}C NMR (125 MHz, CDCl_3): δ = 165.5, 162.4, 161.8, 134.7, 129.1, 123.9, 110.0, 27.9, 21.1 ppm. IR (KBr): $\tilde{\nu}$ = 3521, 3034, 2960, 1786, 1743, 1614, 1467, 1367, 1186, 1135, 1065, 966, 878, 696, 518 cm^{-1} . HRMS (ESI): calcd. for $[\text{M} + \text{Na}]^+$ $\text{C}_{13}\text{H}_{11}\text{NNaO}_4$ 268.0580; found 268.0586.

1,3-Dioxoisindolin-2-yl 4-Nitrobenzoate (3an): White solid. M.p. 198–200 $^{\circ}\text{C}$. ^1H NMR (600 MHz, CDCl_3): δ = 8.39 (s, 4 H), 7.97–7.94 (m, 2 H), 7.87–7.84 (m, 2 H) ppm. ^{13}C NMR (125 MHz, CDCl_3): δ = 161.7, 161.3, 151.6, 135.1, 131.9, 130.8, 128.8, 124.2, 124.0 ppm. IR (KBr): $\tilde{\nu}$ = 3434, 3050, 2926, 1769, 1741, 1608, 1239, 1136, 1053, 1010, 875, 710, 520 cm^{-1} . HRMS (ESI): calcd. for $[\text{M} + \text{Na}]^+$ $\text{C}_{15}\text{H}_8\text{N}_2\text{NaO}_4$ 335.0275; found 335.0272.

1,3-Dioxoisindolin-2-yl 4-Methoxybenzoate (3ao): White solid. M.p. 160–162 $^{\circ}\text{C}$. ^1H NMR (600 MHz, CDCl_3): δ = 8.17 (d, J = 8.4 Hz, 2 H), 7.94–7.93 (m, 2 H), 7.83–7.82 (m, 2 H), 7.02 (d, J = 8.4 Hz, 2 H), 3.92 (s, 3 H) ppm. ^{13}C NMR (125 MHz, CDCl_3): δ = 164.9, 162.4, 162.2, 134.7, 132.9, 129.0, 124.0, 117.3, 114.2, 55.6 ppm. IR (KBr): $\tilde{\nu}$ = 3441, 3024, 2986, 1738, 1605, 1248, 1117, 922, 875, 710, 521 cm^{-1} . HRMS (ESI): calcd. for $[\text{M} + \text{Na}]^+$ $\text{C}_{16}\text{H}_{11}\text{NNaO}_5$ 320.0529; found 320.0529.

1,3-Dioxoisindolin-2-yl 4-Methylbenzoate (3ap): White solid. M.p. 164–165 $^{\circ}\text{C}$. ^1H NMR (600 MHz, CDCl_3): δ = 8.10 (d, J = 7.8 Hz, 2 H), 7.95–7.93 (m, 2 H), 7.84–7.82 (m, 2 H), 7.35 (d, J = 8.4 Hz, 2 H), 2.48 (s, 3 H) ppm. ^{13}C NMR (125 MHz, CDCl_3): δ = 162.8, 162.2, 146.1, 134.8, 130.7, 129.6, 129.1, 124.0, 122.4, 21.9 ppm. IR (KBr): $\tilde{\nu}$ = 3438, 3037, 2932, 1758, 1740, 1607, 1251, 1180, 1119, 1000, 875, 740, 701, 520 cm^{-1} . HRMS (ESI): calcd. for $[\text{M} + \text{Na}]^+$ $\text{C}_{16}\text{H}_{11}\text{NNaO}_4$ 304.0580; found 304.0585.

1,3-Dioxoisindolin-2-yl 2-Naphthoate (3aq): Yellow solid. M.p. 195–197 $^{\circ}\text{C}$. ^1H NMR (600 MHz, CDCl_3): δ = 8.84 (s, 1 H), 8.16 (dd, J = 7.2, 1.2 Hz, 1 H), 8.02 (d, J = 8.4 Hz, 1 H), 7.99–7.94 (m, 4 H), 7.85–7.84 (m, 2 H), 7.71–7.68 (m, 1 H), 7.64–7.61 (m, 1 H) ppm. ^{13}C NMR (125 MHz, CDCl_3): δ = 163.0, 162.2, 136.3, 134.8, 133.0, 132.3, 129.7, 129.4, 129.1, 128.8, 128.0, 127.2, 125.2, 124.1, 122.4 ppm. IR (KBr): $\tilde{\nu}$ = 3445, 3046, 2923, 1770, 1735, 1628, 1177, 1126, 1044, 875, 757, 697, 519 cm^{-1} . HRMS (ESI): calcd. for $[\text{M} + \text{Na}]^+$ $\text{C}_{19}\text{H}_{11}\text{NNaO}_4$ 320.0566; found 320.0580.

1,3-Dioxoisindolin-2-yl Thiophene-2-carboxylate (3ar): White solid. M.p. 145–147 $^{\circ}\text{C}$. ^1H NMR (600 MHz, CDCl_3): δ = 8.09 (d, J = 4.2 Hz, 1 H), 7.94–7.92 (m, 2 H), 7.83–7.81 (m, 2 H), 8.00 (d, J = 4.8 Hz, 1 H), 7.23 (t, J = 4.2 Hz, 1 H) ppm. ^{13}C NMR (125 MHz, CDCl_3): δ = 162.0, 158.3, 136.7, 135.7, 134.9, 128.9, 128.4, 127.0, 124.1 ppm. IR (KBr): $\tilde{\nu}$ = 3442, 3043, 3023, 2926, 1777, 1740, 1639, 1410, 1247, 1187, 1134, 992, 724, 695, 518 cm^{-1} . HRMS (ESI): calcd. for $[\text{M} + \text{Na}]^+$ $\text{C}_{13}\text{H}_7\text{NNaO}_4\text{S}$ 295.9988; found 295.9995.

1,3-Dioxoisindolin-2-yl Furan-2-carboxylate (3as): White solid. M.p. 148–150 $^{\circ}\text{C}$. ^1H NMR (600 MHz, CDCl_3): δ = 7.95–7.93 (m, 2 H), 7.85–7.82 (m, 2 H), 7.78–7.77 (s, 1 H), 7.56 (d, J = 3.6 Hz, 1 H), 6.67–6.66 (m, 1 H) ppm. ^{13}C NMR (125 MHz, CDCl_3): δ = 161.8, 154.5, 148.8, 139.9, 134.9, 128.9, 124.1, 122.2, 112.7 ppm. IR (KBr): $\tilde{\nu}$ = 3444, 3038, 2955, 2924, 1742, 1646, 1466, 1286, 1155, 1081, 1038, 875,

766, 698, 532, 518 cm^{-1} . HRMS (ESI): calcd. for $[\text{M} + \text{Na}]^+$ $\text{C}_{13}\text{H}_7\text{NNaO}_5$ 280.0216; found 280.0230.

1,3-Dioxoisindolin-2-yl Benzoate (3at): White solid. M.p. 168–170 $^{\circ}\text{C}$.^[6c] ^1H NMR (600 MHz, CDCl_3): δ = 8.22 (d, J = 7.8 Hz, 2 H), 7.96–7.93 (m, 2 H), 7.85–7.83 (m, 2 H), 7.72 (t, J = 7.8 Hz, 1 H), 7.56 (t, J = 7.8 Hz, 2 H) ppm. ^{13}C NMR (125 MHz, CDCl_3): δ = 162.8, 162.1, 134.9, 134.8, 130.7, 129.0, 128.9, 125.3, 124.0 ppm. IR (KBr): $\tilde{\nu}$ = 3438, 3040, 3018, 2917, 1773, 1734, 1640, 1237, 1140, 1021, 1007, 875, 697, 520 cm^{-1} . HRMS (ESI): calcd. for $[\text{M} + \text{Na}]^+$ $\text{C}_{15}\text{H}_9\text{NNaO}_4$ 290.0424; found 290.0424.

1,3-Dioxoisindolin-2-yl 4-Fluorobenzoate (3au): White solid. M.p. 189–192 $^{\circ}\text{C}$.^[6c] ^1H NMR (600 MHz, CDCl_3): δ = 8.25–8.22 (m, 2 H), 7.94–7.93 (m, 2 H), 7.83–7.82 (m, 2 H), 7.24–7.21 (m, 2 H) ppm. ^{13}C NMR (125 MHz, CDCl_3): δ = 166.9 ($^1J_{\text{CF}}$ = 213.4 Hz), 162.0, 161.9, 134.9, 133.5 ($^3J_{\text{CF}}$ = 8.1 Hz), 129.0, 124.1, 121.6 ($^4J_{\text{CF}}$ = 2.5 Hz), 116.3 ($^2J_{\text{CF}}$ = 18.5 Hz) ppm. IR (KBr): $\tilde{\nu}$ = 3436, 3053, 3024, 2920, 1773, 1736, 1602, 1239, 1189, 1033, 1009, 875, 700, 520 cm^{-1} . HRMS (ESI): calcd. for $[\text{M} + \text{Na}]^+$ $\text{C}_{15}\text{H}_8\text{FNNaO}_4$ 308.0330; found 308.0316.

1,3-Dioxoisindolin-2-yl 3-Bromobenzoate (3av): White solid. M.p. 169–171 $^{\circ}\text{C}$.^[6c] ^1H NMR (600 MHz, CDCl_3): δ = 8.33 (t, J = 1.8 Hz, 1 H), 8.14–8.12 (m, 1 H), 7.95–7.92 (m, 2 H), 7.84–7.82 (m, 3 H), 7.43 (t, J = 8.4 Hz, 1 H) ppm. ^{13}C NMR (125 MHz, CDCl_3): δ = 161.8, 161.7, 137.9, 134.9, 133.4, 130.5, 129.2, 128.9, 127.2, 124.1, 122.9 ppm. IR (KBr): $\tilde{\nu}$ = 3448, 3094, 3037, 2925, 1774, 1736, 1644, 1185, 1136, 1024, 874, 851, 728, 517 cm^{-1} . HRMS (ESI): calcd. for $[\text{M} + \text{Na}]^+$ $\text{C}_{15}\text{H}_8\text{BrNNaO}_4$ 367.9529; found 367.9508.

1,3-Dioxoisindolin-2-yl 2-Chlorobenzoate (3aw): White solid. M.p. 165–167 $^{\circ}\text{C}$.^[6c] ^1H NMR (600 MHz, CDCl_3): δ = 8.22 (dd, J = 6.6, 1.2 Hz, 1 H), 7.96–7.94 (m, 2 H), 7.85–7.83 (m, 2 H), 7.61–7.57 (m, 2 H), 7.46–7.44 (m, 1 H) ppm. ^{13}C NMR (125 MHz, CDCl_3): δ = 161.9, 161.2, 134.9, 134.6, 132.5, 131.7, 129.0, 126.9, 124.8, 124.1, 123.6 ppm. IR (KBr): $\tilde{\nu}$ = 3419, 3068, 2958, 1777, 1751, 1638, 1588, 1229, 1185, 1137, 1075, 996, 875, 694, 518 cm^{-1} . HRMS (ESI): calcd. for $[\text{M} + \text{Na}]^+$ $\text{C}_{15}\text{H}_8\text{ClNNaO}_4$ 324.0034; found 324.0050.

1,3-Dioxoisindolin-2-yl 4-Cyanobenzoate (3ax): White solid. M.p. 218–220 $^{\circ}\text{C}$.^[6c] ^1H NMR (600 MHz, CDCl_3): δ = 8.33 (d, J = 8.4 Hz, 2 H), 7.98–7.96 (m, 2 H), 7.89–7.86 (m, 4 H) ppm. ^{13}C NMR (125 MHz, CDCl_3): δ = 161.7, 161.5, 135.0, 132.6, 131.1, 129.2, 128.8, 124.2, 118.3, 117.4 ppm. IR (KBr): $\tilde{\nu}$ = 3438, 3043, 3020, 2928, 2235, 1776, 1744, 1602, 1229, 1169, 1053, 1000, 876, 696, 518 cm^{-1} . HRMS (ESI): calcd. for $[\text{M} + \text{Na}]^+$ $\text{C}_{16}\text{H}_8\text{N}_2\text{NaO}_4$ 315.0381; found 315.0388.

1,3-Dioxoisindolin-2-yl Biphenyl-4-carboxylate (3ay): White solid. M.p. 208–210 $^{\circ}\text{C}$. ^1H NMR (600 MHz, CDCl_3): δ = 8.29 (d, J = 8.4 Hz, 2 H), 7.98–7.95 (m, 2 H), 7.85–7.84 (m, 2 H), 7.79 (d, J = 8.4 Hz, 2 H), 7.69–7.68 (m, 2 H), 7.53 (t, J = 7.2 Hz, 2 H), 7.47–7.45 (m, 2 H) ppm. ^{13}C NMR (125 MHz, CDCl_3): δ = 162.8, 162.1, 147.7, 139.5, 134.8, 131.2, 129.1, 128.7, 127.5, 127.4, 124.1, 123.9 ppm. IR (KBr): $\tilde{\nu}$ = 3418, 3069, 3021, 2923, 1775, 1744, 1607, 1376, 1254, 1187, 1141, 1035, 1001, 876, 740, 694, 518 cm^{-1} . HRMS (ESI): calcd. for $[\text{M} + \text{Na}]^+$ $\text{C}_{21}\text{H}_{13}\text{NNaO}_4$ 366.0737; found 366.0734.

1,3-Dioxoisindolin-2-yl Methyl Terephthalate (3az): White solid. M.p. 180–182 $^{\circ}\text{C}$.^[6c] ^1H NMR (600 MHz, CDCl_3): δ = 8.29 (d, J = 8.4 Hz, 2 H), 8.21 (d, J = 8.4 Hz, 2 H), 7.97–7.94 (m, 2 H), 7.86–7.84 (m, 2 H), 4.00 (s, 3 H) ppm. ^{13}C NMR (125 MHz, CDCl_3): δ = 165.8, 162.2, 161.9, 135.6, 134.9, 130.6, 129.9, 129.0, 128.9, 124.1, 52.7 ppm. IR (KBr): $\tilde{\nu}$ = 3422, 3094, 2961, 1772, 1739, 1637, 1280, 1237, 1138, 1103, 1043, 1010, 875, 696, 519 cm^{-1} . HRMS (ESI): calcd. for $[\text{M} + \text{Na}]^+$ $\text{C}_{17}\text{H}_{11}\text{NNaO}_6$ 348.0468; found 348.0479.

1,3-Dioxoisindolin-2-yl 1-Naphthoate (3aaa): White solid. M.p. 181–183 $^{\circ}\text{C}$.^[6c] ^1H NMR (600 MHz, CDCl_3): δ = 8.87 (d, J = 8.4 Hz, 1

H), 8.57 (dd, J = 6.6, 0.6 Hz, 1 H), 8.19 (d, J = 7.8 Hz, 1 H), 7.99–7.95 (m, 3 H), 7.85–7.84 (m, 2 H), 7.70–7.68 (m, 1 H), 7.63–7.60 (m, 2 H) ppm. ^{13}C NMR (125 MHz, CDCl_3): δ = 163.1, 162.3, 135.6, 134.8, 133.8, 131.9, 131.5, 129.1, 128.8, 128.7, 126.8, 125.4, 124.5, 124.1, 121.7 ppm. IR (KBr): $\tilde{\nu}$ = 3434, 3046, 2917, 1772, 1743, 1365, 1136, 1096, 978, 773, 695, 520 cm^{-1} . HRMS (ESI): calcd. for $[\text{M} + \text{Na}]^+$ $\text{C}_{20}\text{H}_{27}\text{NNaO}_4$ 340.0580; found 340.0569.

1,3-Dioxoisindolin-2-yl Dodecanoate (3aab): Oil. ^1H NMR (600 MHz, CDCl_3): δ = 7.89–7.88 (m, 2 H), 7.80–7.78 (m, 2 H), 2.66 (t, J = 7.2 Hz, 2 H), 1.81–1.76 (m, 2 H), 1.47–1.42 (m, 2 H), 1.34–1.27 (m, 14 H), 0.88 (t, J = 7.2 Hz, 3 H) ppm. ^{13}C NMR (125 MHz, CDCl_3): δ = 169.7, 162.0, 134.7, 129.0, 123.9, 31.9, 31.0, 29.58, 29.57, 29.4, 29.3, 29.1, 28.8, 24.7, 22.7, 14.1 ppm. IR (KBr): $\tilde{\nu}$ = 3427, 3091, 3062, 2954, 2920, 1789, 1744, 1375, 1185, 1143, 962, 880, 696, 521 cm^{-1} . HRMS (ESI): calcd. for $[\text{M} + \text{Na}]^+$ $\text{C}_{20}\text{H}_{27}\text{NNaO}_4$ 368.1832; found 368.1815.

2,5-Dioxopyrrolidin-1-yl 4-Fluorobenzoate (3ba): White solid. M.p. 106–108 $^{\circ}\text{C}$.^[6b] ^1H NMR (600 MHz, CDCl_3): δ = 8.20–8.17 (m, 2 H), 7.21 (t, J = 8.4 Hz, 2 H), 2.93 (s, 4 H) ppm. ^{13}C NMR (125 MHz, CDCl_3): δ = 169.3, 166.9 ($^1J_{\text{CF}}$ = 213.5 Hz), 161.0, 133.4 ($^3J_{\text{CF}}$ = 8.1 Hz), 121.4 ($^4J_{\text{CF}}$ = 2.5 Hz), 116.3 ($^2J_{\text{CF}}$ = 18.5 Hz), 25.7 ppm. IR (KBr): $\tilde{\nu}$ = 3430, 3074, 3023, 2964, 1778, 1730, 1598, 1400, 1085, 1056, 990, 846, 747, 530 cm^{-1} . HRMS (ESI): calcd. for $[\text{M} + \text{Na}]^+$ $\text{C}_{11}\text{H}_8\text{FNNaO}_4$ 260.0330; found 260.0320.

2,5-Dioxopyrrolidin-1-yl 4-Chlorobenzoate (3bb): White solid. M.p. 205–207 $^{\circ}\text{C}$.^[6c] ^1H NMR (600 MHz, CDCl_3): δ = 8.09 (d, J = 8.4 Hz, 2 H), 7.56 (d, J = 8.4 Hz, 2 H), 2.93 (s, 4 H) ppm. ^{13}C NMR (125 MHz, CDCl_3): δ = 169.2, 161.2, 141.7, 131.9, 129.4, 123.6, 25.7 ppm. IR (KBr): $\tilde{\nu}$ = 3442, 3080, 3070, 2967, 1770, 1744, 1567, 1211, 1070, 1060, 991, 848, 730, 520 cm^{-1} . HRMS (ESI): calcd. for $[\text{M} + \text{Na}]^+$ $\text{C}_{11}\text{H}_8\text{ClNNaO}_4$ 276.0034; found 276.0018.

2,5-Dioxopyrrolidin-1-yl 4-Bromobenzoate (3bc): White solid. M.p. 206–208 $^{\circ}\text{C}$.^[6f] ^1H NMR (600 MHz, CDCl_3): δ = 8.01 (d, J = 8.4 Hz, 2 H), 7.69 (d, J = 8.4 Hz, 2 H), 2.93 (s, 4 H) ppm. ^{13}C NMR (125 MHz, CDCl_3): δ = 169.1, 161.3, 132.4, 131.9, 130.5, 124.0, 25.7 ppm. IR (KBr): $\tilde{\nu}$ = 3432, 3084, 3069, 2974, 1773, 1727, 1588, 1406, 1211, 1080, 1066, 991, 848, 740, 531 cm^{-1} . HRMS (ESI): calcd. for $[\text{M} + \text{Na}]^+$ $\text{C}_{11}\text{H}_8\text{BrNNaO}_4$ 319.9529; found 319.9515.

2,5-Dioxopyrrolidin-1-yl 4-Nitrobenzoate (3bd): White solid. M.p. 200–203 $^{\circ}\text{C}$.^[6b] ^1H NMR (600 MHz, CDCl_3): δ = 8.40 (d, J = 9.0 Hz, 2 H), 8.35 (d, J = 9.0 Hz, 2 H), 2.97 (s, 4 H) ppm. ^{13}C NMR (125 MHz, CDCl_3): δ = 168.8, 160.3, 151.5, 131.8, 130.6, 124.0, 25.7 ppm. IR (KBr): $\tilde{\nu}$ = 3445, 3094, 3062, 2983, 1771, 1726, 1588, 1210, 1080, 1066, 991, 847, 740, 530 cm^{-1} . HRMS (ESI): calcd. for $[\text{M} + \text{Na}]^+$ $\text{C}_{11}\text{H}_8\text{NNaO}_6$ 287.0380; found 287.0392.

2,5-Dioxopyrrolidin-1-yl 4-Methoxybenzoate (3be): White solid. M.p. 149–151 $^{\circ}\text{C}$.^[6b] ^1H NMR (600 MHz, CDCl_3): δ = 8.11 (d, J = 9.0 Hz, 2 H), 7.00 (d, J = 9.0 Hz, 2 H), 3.91 (s, 3 H), 2.91 (s, 4 H) ppm. ^{13}C NMR (125 MHz, CDCl_3): δ = 169.6, 164.9, 161.5, 132.9, 117.1, 114.2, 55.6, 25.7 ppm. IR (KBr): $\tilde{\nu}$ = 3445, 3081, 2974, 2932, 1766, 1736, 1602, 1514, 1428, 1217, 1183, 1074, 984, 848, 757, 595, 517 cm^{-1} . HRMS (ESI): calcd. for $[\text{M} + \text{Na}]^+$ $\text{C}_{12}\text{H}_{11}\text{NNaO}_5$ 272.0529; found 272.0532.

2,5-Dioxopyrrolidin-1-yl 3-Methoxybenzoate (3bf): White solid. M.p. 102–104 $^{\circ}\text{C}$.^[6c] ^1H NMR (600 MHz, CDCl_3): δ = 7.76 (dd, J = 7.2, 0.6 Hz, 1 H), 7.62 (t, J = 1.2 Hz, 1 H), 7.43 (t, J = 8.4 Hz, 1 H), 7.24–7.22 (m, 1 H), 3.87 (s, 3 H), 2.93 (s, 4 H) ppm. ^{13}C NMR (125 MHz, CDCl_3): δ = 169.4, 161.8, 159.7, 130.0, 126.2, 123.0, 121.7, 114.6, 55.6, 25.7 ppm. IR (KBr): $\tilde{\nu}$ = 3445, 3094, 3049, 2963, 1770, 1744, 1570, 1426, 1201, 1089, 1076, 991, 858, 740, 696, 529 cm^{-1} . HRMS (ESI): calcd. for $[\text{M} + \text{Na}]^+$ $\text{C}_{12}\text{H}_{11}\text{NNaO}_5$ 272.0529; found 272.0516.

2,5-Dioxopyrrolidin-1-yl 2-Methoxybenzoate (3bg): White solid. M.p. 174–176 °C.^[6c] ¹H NMR (600 MHz, CDCl₃): δ = 8.08 (dd, *J* = 6.0, 1.8 Hz, 1 H), 7.64–7.61 (m, 1 H), 7.06 (t, *J* = 7.8 Hz, 2 H), 3.95 (s, 3 H), 2.92 (s, 4 H) ppm. ¹³C NMR (125 MHz, CDCl₃): δ = 169.5, 160.7, 160.3, 136.0, 132.7, 120.3, 114.0, 112.2, 56.1, 25.7 ppm. IR (KBr): ν̄ = 3434, 3074, 3029, 2983, 1768, 1740, 1470, 1231, 1100, 1076, 997, 898, 747, 700, 530 cm⁻¹. HRMS (ESI): calcd. for [M + Na]⁺ C₁₂H₁₁NNaO₅ 272.0529; found 272.0539.

2,5-dioxopyrrolidin-1-yl 4-Methylbenzoate (3bh): White solid. M.p. 173–175 °C.^[6c] ¹H NMR (600 MHz, CDCl₃): δ = 8.04 (d, *J* = 8.4 Hz, 2 H), 7.33 (d, *J* = 8.4 Hz, 2 H), 2.91 (s, 4 H), 2.46 (s, 3 H) ppm. ¹³C NMR (125 MHz, CDCl₃): δ = 169.4, 161.9, 146.1, 130.6, 129.6, 122.3, 25.7, 21.9 ppm. IR (KBr): ν̄ = 3444, 3094, 3039, 2983, 1771, 1743, 1577, 1436, 1200, 1089, 1046, 1000, 878, 749, 696, 529 cm⁻¹. HRMS (ESI): calcd. for [M + Na]⁺ C₁₂H₁₁NNaO₄ 256.0580; found 256.0590.

2,5-dioxopyrrolidin-1-yl Methyl Terephthalate (3bi): White solid. M.p. 173–174 °C.^[6c] ¹H NMR (600 MHz, CDCl₃): δ = 8.22 (d, *J* = 7.2 Hz, 2 H), 8.19 (d, *J* = 7.2 Hz, 2 H), 3.99 (s, 3 H), 2.94 (s, 4 H) ppm. ¹³C NMR (125 MHz, CDCl₃): δ = 169.1, 165.8, 161.2, 135.6, 130.6, 129.9, 128.9, 52.7, 25.7 ppm. IR (KBr): ν̄ = 3435, 3008, 2951, 1732, 1710, 1288, 1235, 1207, 1110, 1067, 1001, 845, 720, 529 cm⁻¹. HRMS (ESI): calcd. for [M + Na]⁺ C₁₃H₁₁NNaO₆ 300.0479; found 300.0490.

Acknowledgments

We are grateful to the National Natural Science Foundation of China (grant No. 21576239) for financial support. We thank Dr. Qi Shuai at Zhejiang University of Technology for helpful discussions.

Keywords: Earth-abundant metals · Oxidation · Cross-dehydrogenative coupling · Esters · Aldehydes · Alcohols

- [1] a) K. Ishihara, *Tetrahedron* **2009**, *65*, 1085; b) J. Otera, *Esterification: Methods, Reactions, and Applications*, Wiley-VCH, Weinheim, **2003**; c) T. W. Greene, P. G. M. Wuts, *Protective Groups in Organic Synthesis*, Wiley-VCH, New York, **1991**; d) D. V. Rosato, M. V. Rosato, *Plastic Product Material and Process Selection Handbook*, Elsevier, New York, **2004**; e) W. Riemenschneider, H. M. Bolt, "Esters, Organic" in *Ullmann's Encyclopedia of Industrial Chemistry*, Wiley-VCH, Weinheim, **2005**.
- [2] a) C. A. G. N. Montalbetti, V. Falque, *Tetrahedron* **2005**, *61*, 10827; b) E. Valeur, M. Bradley, *Chem. Soc. Rev.* **2009**, *38*, 606; c) V. R. Pattabiraman, J. W. Bode, *Nature* **2011**, *480*, 471; d) S. K. Santra, A. Banerjee, S. Rajamanickam, N. Khatun, B. K. Patel, *Chem. Commun.* **2016**, *52*, 4501; e) S. K. Rout, S. Guin, W. Ali, A. Gogoi, B. K. Patel, *Org. Lett.* **2014**, *16*, 3086; f) S. K. Rout, S. Guin, A. Gogoi, G. Majji, B. K. Patel, *Org. Lett.* **2014**, *16*, 1614; g) G. Majji, S. K. Rout, S. Rajamanickam, S. Guin, B. K. Patel, *Org. Biomol. Chem.* **2016**, *14*, 8178; h) A. Banerjee, S. Sarkar, B. K. Patel, *Org. Biomol. Chem.* **2017**, *15*, 505; i) S. K. Rout, S. Guin, A. Banerjee, N. Khatun, A. Gogoi, B. K. Patel, *Org. Lett.* **2013**, *15*, 4016; j) S. K. Rout, S. Guin, K. K. Ghara, A. Banerjee, B. K. Patel, *Org. Lett.* **2012**, *14*, 3982; k) C. Liu, S. Tang, L. W. Zheng, D. Liu, H. Zhang, A. W. Lei, *Angew. Chem. Int. Ed.* **2012**, *51*, 5662; *Angew. Chem.* **2012**, *124*, 5760; l) E. K. Kekeli, W. Christian, *Chem. Eur. J.* **2008**, *14*, 6302; m) S. Gowrisankar, H. Neumann, M. Beller, *Angew. Chem. Int. Ed.* **2011**, *50*, 5139; *Angew. Chem.* **2011**, *123*, 5245; n) Y. H. Zhang, Ch. J. Li, *Angew. Chem. Int. Ed.* **2006**, *45*, 1949; *Angew. Chem.* **2006**, *118*, 1983.

- [3] a) G. H. L. Nefkens, G. I. Tesser, *J. Am. Chem. Soc.* **1961**, *83*, 1263; b) G. W. Anderson, J. E. Zimmerman, F. M. Callahan, *J. Am. Chem. Soc.* **1964**, *86*, 1839; c) G. W. Cline, S. B. Hanna, *J. Am. Chem. Soc.* **1987**, *109*, 3087; d) J. M. Takacs, S. D. Schroeder, J. Han, M. Gifford, X. T. Jiang, T. Saleh, S. Vayalakkada, A. H. Yap, *Org. Lett.* **2003**, *5*, 3595; e) M. M. Huihui, J. A. C. Kierra, M. Zulema, M. O. Astrid, M. S. Amanda, A. J. Keywan, A. D. Tarah, K. Seoyoung, K. G. A. Laura, J. W. Daniel, *J. Am. Chem. Soc.* **2016**, *138*, 5016; f) O. K. Mikhail, R. J. Elizabeth, *Angew. Chem. Int. Ed.* **2016**, *55*, 11340; *Angew. Chem.* **2016**, *128*, 11510.
- [4] a) E. Grochowski, J. Jurczak, *Synthesis* **1977**, 277; b) H. Ogura, T. Kobayashi, K. Shimizu, K. Kawabe, K. Takeda, *Tetrahedron Lett.* **1979**, *20*, 4745; c) S. Kim, K. Y. Ko, *J. Chem. Soc., Chem. Commun.* **1985**, 473; d) P. Pçchlauer, W. Hendel, *Tetrahedron* **1998**, *54*, 3489; e) M. Kim, K. J. Han, *Synth. Commun.* **2009**, *39*, 4467.
- [5] a) C. S. Yeung, V. M. Dong, *Chem. Rev.* **2011**, *111*, 1215; b) C. J. Scheuermann, *Chem. Asian J.* **2010**, *5*, 436; c) J. LeBras, J. Muzart, *Chem. Rev.* **2011**, *111*, 1170.
- [6] a) B. Tan, N. Toda, C. F. Barbas III, *Angew. Chem. Int. Ed.* **2012**, *51*, 12538; *Angew. Chem.* **2012**, *124*, 12706; b) M. Dinda, C. Bose, T. Ghosh, S. Maity, *RSC Adv.* **2015**, *5*, 44928; c) Y. H. Lv, K. Sun, W. Y. Pu, S. K. Mao, G. Li, J. J. Niu, Q. Chen, T. T. Wang, *RSC Adv.* **2016**, *6*, 93486; d) Z. Guo, X. Jiang, C. Jin, J. Zhou, B. Sun, W. Su, *Synlett* **2017**, 1321; e) A. Schulze, A. Giannis, *Adv. Synth. Catal.* **2004**, *346*, 252; f) X. H. Xu, P. P. Li, Y. Y. Huang, Ch. Tong, Y. Y. Yan, Y. Y. Xie, *Tetrahedron Lett.* **2017**, *58*, 1742.
- [7] G. Wang, Q. Y. Yu, J. Wang, S. Wang, S. Y. Chen, X. Q. Yu, *RSC Adv.* **2013**, *3*, 21306.
- [8] S. Enthaler, K. Junge, M. Beller, *Angew. Chem. Int. Ed.* **2008**, *47*, 3317; *Angew. Chem.* **2008**, *120*, 3363.
- [9] a) C. Bolm, J. Legros, J. Le Pailh, L. Zani, *Chem. Rev.* **2004**, *104*, 6217; b) C. L. Sun, B. J. Li, Z. J. Shi, *Chem. Rev.* **2011**, *111*, 1293; c) J. Yang, T. D. Tilley, *Angew. Chem. Int. Ed.* **2010**, *49*, 10186; *Angew. Chem.* **2010**, *122*, 10384; d) S. Ito, T. Itoh, M. Nakamura, *Angew. Chem. Int. Ed.* **2011**, *50*, 454; *Angew. Chem.* **2011**, *123*, 474; e) R. B. Bedford, M. A. Hall, G. R. Hodges, M. Huwe, M. C. Wilkinson, *Chem. Commun.* **2009**, 6430.
- [10] a) Ch. X. Miao, H. Q. Zhao, Q. Y. Zhao, Ch. G. Xia, W. Sun, *Catal. Sci. Technol.* **2016**, *6*, 1378; b) S. Ma, J. Liu, S. Li, B. Chen, J. Cheng, J. Kuang, Y. Liu, B. Wan, Y. Wang, J. Ye, Q. Yu, W. Yuan, S. Yu, *Adv. Synth. Catal.* **2011**, *353*, 1005; c) B. Raghunath, S. Dinabandhu, T. Punniyamurthy, *Org. Biomol. Chem.* **2016**, *14*, 3246; d) J. Z. Zhang, Y. Tang, *Adv. Synth. Catal.* **2016**, *358*, 752; e) S. Pradeep, K. A. Santhosh, T. Punniyamurthy, *J. Org. Chem.* **2015**, *80*, 8245; f) T. Tsuyoshi, I. Hiroyuki, *Org. Lett.* **2010**, *12*, 124; g) T. Naveen, S. Maity, U. Sharma, D. Maiti, *J. Org. Chem.* **2013**, *78*, 5949; h) R. Bag, P. Bhusan De, S. Pradhan, T. Punniyamurthy, *Eur. J. Org. Chem.* **2017**, 5424; i) R. Bag, D. Sar, T. Punniyamurthy, *ACS Omega* **2017**, *2*, 6278.
- [11] a) W. D. Hill Jr., *Inorg. Chim. Acta* **1986**, *121*, 33; b) K. Wierzchorek-Ciurawa, A. J. Kozak, *J. Therm. Anal. Calorim.* **1999**, *58*, 647.
- [12] a) Y. Ishii, S. Sakaguchi, Y. Nishiwaki, T. Kitamura, *Angew. Chem. Int. Ed.* **2001**, *40*, 222; *Angew. Chem.* **2001**, *113*, 228; b) N. Sawatari, S. Sakaguchi, Y. Ishii, *Tetrahedron Lett.* **2003**, *44*, 2053; c) S. Sakaguchi, M. Eikawa, Y. Ishii, *Tetrahedron Lett.* **1997**, *38*, 7075; d) M. Eikawa, S. Sakaguchi, Y. Ishii, *J. Org. Chem.* **1999**, *64*, 4676; e) S. Sakaguchi, T. Hirabayashi, Y. Ishii, *Chem. Commun.* **2002**, 516; f) T. Hirabayashi, S. Sakaguchi, Y. Ishii, *Angew. Chem. Int. Ed.* **2004**, *43*, 1120; *Angew. Chem.* **2004**, *116*, 1140.
- [13] a) X. L. Wang, R. H. Liu, Y. Jin, X. M. Liang, *Chem. Eur. J.* **2008**, *14*, 2679; b) B. Karimi, A. Biglari, J. M. Clark, W. Budarin, *Angew. Chem. Int. Ed.* **2007**, *46*, 7210; *Angew. Chem.* **2007**, *119*, 7348.
- [14] a) C. Chatgililoglu, D. Crich, M. Komatsu, I. Ryu, *Chem. Rev.* **1999**, *99*, 1991; b) S. Tsujimoto, S. Sakaguchi, Y. Ishii, *Tetrahedron Lett.* **2003**, *44*, 5601; c) G. A. DiLabio, K. U. Ingold, M. D. Roydhouse, J. C. Walton, *Org. Lett.* **2004**, *6*, 4319; d) M. Conte, H. Miyamura, S. Kobayashi, V. Chechik, *Chem. Commun.* **2010**, 46, 145; e) Z. J. Liu, J. Zhang, S. L. Chen, E. B. Shi, Y. Xu, X. B. Wan, *Angew. Chem. Int. Ed.* **2012**, *51*, 3231; *Angew. Chem.* **2012**, *124*, 3285; f) Y. F. Liang, X. Wang, C. Tang, T. Jian, Z. Liu, N. Jiao, *Chem. Commun.* **2016**, *52*, 1416.

Received: October 9, 2017